

| Assessment | Test | Domain        | Skill                                                                         | Difficulty                                        |
|------------|------|---------------|-------------------------------------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear equations in one variable and systems of equations in two variables | Hard <div><div></div><div></div><div></div></div> |

Question

In the  $xy$ -plane, a line with equation  $2y = 4.5$  intersects a parabola at exactly one point. If the parabola has equation  $y = -4x^2 + bx$ , where  $b$  is a positive constant, what is the value of  $b$ ?

Correct Answer: 6

Rationale

The correct answer is **6**. It's given that a line with equation  $2y = 4.5$  intersects a parabola with equation  $y = -4x^2 + bx$ , where  $b$  is a positive constant, at exactly one point in the  $xy$ -plane. It follows that the system of equations consisting of  $2y = 4.5$  and  $y = -4x^2 + bx$  has exactly one solution. Dividing both sides of the equation of the line by **2** yields  $y = 2.25$ . Substituting **2.25** for  $y$  in the equation of the parabola yields  $2.25 = -4x^2 + bx$ . Adding  $4x^2$  and subtracting  $bx$  from both sides of this equation yields  $4x^2 - bx + 2.25 = 0$ . A quadratic equation in the form of  $ax^2 + bx + c = 0$ , where  $a$ ,  $b$ , and  $c$  are constants, has exactly one solution when the discriminant,  $b^2 - 4ac$ , is equal to zero. Substituting **4** for  $a$  and **2.25** for  $c$  in the expression  $b^2 - 4ac$  and setting this expression equal to **0** yields  $b^2 - 4(4)(2.25) = 0$ , or  $b^2 - 36 = 0$ . Adding **36** to each side of this equation yields  $b^2 = 36$ . Taking the square root of each side of this equation yields  $b = \pm 6$ . It's given that  $b$  is positive, so the value of  $b$  is **6**.